

GIACOMO MARCIANI

SENIOR SOFTWARE ENGINEER @ Amazon Web Services

Building High Performance Computing in the Cloud



LOCATION

Boston, MA, United States

CONTACTS

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gmarciani (LinkedIn)

gmarciani (GitHub)

INTRODUCTION

I am a Senior Software Engineer with 8+ years at AWS, specializing in High Performance Computing in cloud environments and large-scale infrastructure automation. I own the full lifecycle end-to-end: strategy, design, implementation, DevOps, and on-call operations, with experience leading technical teams and shaping product direction.

I have contributed to AWS ParallelCluster, NICE EnginFrame, Elastic Graphics, and Elastic Inference. My background spans software engineering, distributed systems and big data. Before AWS, I founded the first ACM Student Chapter in Rome, co-founded a startup, and taught computer science at university. I also led research projects that resulted in publications in scientific journals and presentations at international conferences.

What makes me effective as an engineer and a leader is a relentless curiosity; a drive to keep improving how I build, how I lead, and how I show up for my team.

WORK EXPERIENCE

Sr Software Engineer

Amazon Web Services, 2021-Present

Technical leader of AWS ParallelCluster. My responsibilities span the software lifecycle end-to-end: strategy, design, implementation, observability, on-call incident response, mentoring, and hiring.

Software Engineer II

Amazon Web Services, 2019-2021

Worked on Elastic Graphics, Elastic Inference, and NICE EnginFrame. On EnginFrame, I contributed mainly to the build system, CI/CD automation, and integration work to enable the portal to interact with both on-premises and cloud-based clusters. Starting in 2021, I took on AWS ParallelCluster, where I became the team's de facto technical leader. My key contributions include Multi-User environment, Dynamic File System Mounting, support for GPU-intensive instance types, enabling ParallelCluster in US isolated regions, Terraform integration, ParallelCluster UI, serving as Security Guardian, driving roadmap and releases, and resolving high-severity escalations for key customers. I also led the complex shutdown of Elastic Graphics and Elastic Inference, navigating deep integrations with EC2/ECS.

Software Engineer I

Amazon Web Services, 2018-2019

Worked on Elastic Graphics and Elastic Inference, which attached cost-effective GPU acceleration to EC2 instances and other AWS services. I contributed to the launch of Elastic Inference at Re:Invent 2018 and owned the metering and billing component for compute resources.

R&D Intern

Accenture, 2017

Designed a solution for the elastic auto-scaling of containerized applications using Reinforcement Learning in Kubernetes.

Chairman, Founder

ACM Rome Tor Vergata, 2016-2018

Founded the ACM Student Chapter at University of Rome Tor Vergata, the first in the city. Led the team and fostered connections between academia and industry.

Teaching Assistant

University of Rome Tor Vergata, 2016-2018

Tutored undergraduate students and graded exams for Foundations of Computer Science and Algorithms Engineering.

Mathematics Tutor

Freelance, 2011-2017

Gave private math lessons to secondary and high school students. Several of my students went on to study engineering.

Management Engineer, Co-Founder

Sbaam.com, e-Commerce, 2011-2012

Co-founded the startup, making it a finalist in the InnovActionLab startups contest. Responsible for business plan development and investor pitches.

MAIN PROJECTS

AWS ParallelCluster and ParallelCluster UI

Amazon Web Services, 2021-Present | HPC

Open-source platform to automate deployment and lifecycle management of large-scale HPC clusters on AWS, with millions of dollars in revenue and major enterprise customers.

NICE EnginFrame

Amazon Web Services, 2021 | HPC

Web application to submit HPC workloads to both on-premises and AWS clusters.

AWS Elastic Graphics and Elastic Inference

Amazon Web Services, 2018-2023 | Web Service

Services to attach GPU-powered acceleration for OpenGL and ML workloads to EC2, ECS and Sagemaker to reduce the cost of running graphics and machine learning applications.

Crime Graph

Big Data, IEEE FiCloud, 2017 | Java, Kafka, Flink, Neo4j

Data stream processing application for real-time analysis of evolving criminal networks, uncovering hidden and latent relationships.

PyDES

Performance Modeling, 2019 | Python

Discrete-event simulation suite for performance modeling of cloud computing systems, featuring batch-means analysis, Markov chain solvers, and multi-stream pseudo-random number generation.

OntoQA

Artificial Intelligence, 2017 | NLP, Java, Protege

Question&Answer web application leveraging ontology-driven natural language processing.

Godot

App for Transports, 2013 | Java, Android

Mobile app to address issues caused by illegal parking, leveraging QR codes and NFC.

See more on my [GitHub](#)

PUBLICATIONS

A data streaming approach to link mining in criminal networks

5th IEEE International Conference on Future IoT (FiCloud 2017)

Real-time analysis of social networks leveraging Flink

10th ACM International Conference on Distributed and Event-Based Systems (DEBS 2016)

Marciani Normal Form of context-free grammars

Arxiv (2016)

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EDUCATION

MSc Software Engineering and Data Science

University of Rome Tor Vergata, 2015-2018

Grad Avg 29/30 - GPA 4/4 *

BSc Software Engineering

University of Rome Tor Vergata, 2011-2015

Grad 110/110 cum laude

BSc Management Engineering

University of Rome Tor Vergata, 2009-2011

Interrupted

MAIN SKILLS

Cloud Amazon Web Services, Digital Ocean, Railway

HPC ParallelCluster, EnginFrame, Slurm, Nvidia GPUs, Cuda, MPI, Lustre, EFA

Big Data Flink, Hadoop Distributed File System, Hadoop MapReduce, Pig, Hive, Kafka, Flume, Elasticsearch, Kibana, Grafana, Kubernetes

Web Spring, Nginx, Tomcat, React, NodeJS, HTML, CSS, Sass, Postman, Hugo, Wordpress

Automation CloudFormation, Terraform, Chef

Data SQL, NoSQL, Graph, Caching

Languages Python (primary), Java, Kotlin, Ruby, Javascript, C, Cuda, Bash, HTML, OWL, SPARQL, AMPL, MIPS, Latex

Platforms Linux, OSX, Docker

Applied Math Matlab, AMPL, IBM Cplex

Process Agile, SCRUM

HONOURS

Lecturer at Master CESMA

University of Rome Tor Vergata, 2016

Finalist at ACM DEBS Grand Challenge

10th ACM International Conference on Distributed Event Based Systems, 2016

Speaker at InnovAction Lab Grand Challenge

LVenture Groups, 2013

Best Student in Computer Architectures

University of Rome Tor Vergata, 2012

Alumno of InnovAction Lab

LVenture Groups, 2011

Counselor of Engineering

University of Rome Tor Vergata, 2010

LANGUAGES

Italian

Native

English

Proficient